LETTER

Evaluation design decides the winner: spatial and acquisition leakage in few-label hyperspectral classification

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ABSTRACT
Comparisons between deep models and gradient boosting for hyperspectral classification with few labels depend on how the data are split and prepared. We compare LightGBM with three transformers and the 3-D CNN HybridSN under leaky and controlled protocols, using five paired seeds and false-discovery-rate control. On Indian Pines, a random pixel split puts HybridSN 17–20 percentage points (pp) ahead of LightGBM from 5% labelled pixels; on spatially disjoint tiles no deep model is significantly better than LightGBM. On Pavia University and Salinas, HybridSN stays ahead under disjoint tiles (+3 to +9 pp over LightGBM), and a random forest is at least as strong a baseline as LightGBM. On laboratory images of Indian soils, 38–72% of patches show only the sample tray, and a classifier trained on tray patches alone reaches 87% accuracy (chance 33%). Removing the tray and normalising brightness lifts every model to $\geq$94% with one labelled sample per class and removes most differences between them. Spatially disjoint splits, background masking and a tray-only check should be standard before models are ranked.

KEYWORDS
Hyperspectral imaging; data leakage; spatially disjoint split; LightGBM; transformer; soil classification

1. Introduction

Hyperspectral imaging (HSI) records hundreds of narrow spectral bands and is used for agricultural and soil surveys (Bioucas-Dias et al. 2013). Labels are expensive, so classifiers are usually trained on a small share of the pixels, and the choice between deep models (Zhong et al. 2018; Roy et al. 2020; Hong et al. 2021, 2022) and simpler learners such as gradient-boosted trees (Ke et al. 2017; Grinsztajn, Oyallon, and Varoquaux 2022) matters in practice.

Such comparisons can be distorted in two ways. *Spatial leakage*: with a random pixel split, training and test patches overlap, which favours patch-based models (Nalepa,

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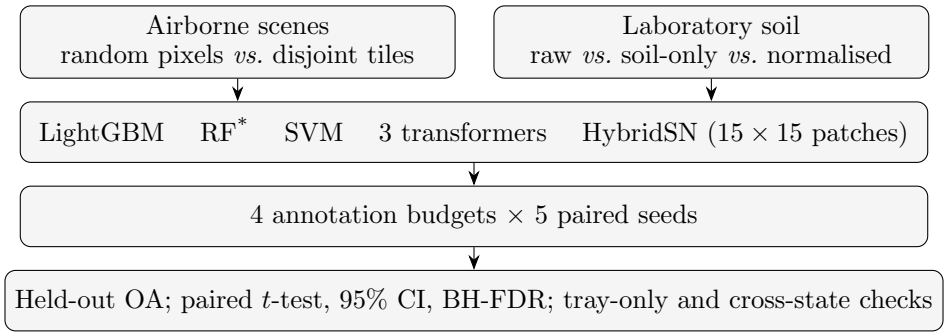


Figure 1. Experimental design. Each dataset is evaluated under a leaky and a controlled protocol with the same models, seeds and statistics. *Random forest on the airborne scenes only.

Myller, and Kawulok 2019). *Acquisition leakage*: in laboratory imaging, background pixels and session-specific illumination can carry the class label, so a model may recognise the imaging session rather than the material. The first effect is known but still common; the second is rarely checked.

We measure how much each effect changes the ranking of five learners at small annotation budgets. Our contributions are: (i) the same models, seeds and statistics under a random and a spatially disjoint split of Indian Pines, and a disjoint benchmark on two further airborne scenes; (ii) for a laboratory soil dataset, a comparison of raw, soil-only and brightness-normalised patches, a tray-only leakage check and a moist/dry cross-state test; and (iii) a short checklist for few-label HSI comparisons.

2. Data

Airborne scenes. Indian Pines (AVIRIS; 145 × 145 pixels, 200 bands after water-band removal, 16 classes, 10 249 labelled pixels; Baumgardner, Biehl, and Landgrebe 2015), Pavia University (ROSIS; 610 × 340 pixels, 103 bands, 9 classes, 42 776 labelled pixels; Gamba 2004) and Salinas (AVIRIS; 512 × 217 pixels, 204 bands, 16 classes, 54 129 labelled pixels; Grupo de Inteligencia Computacional 2021). Indian Pines band centres are taken from the 1992 AVIRIS calibration (Baumgardner, Biehl, and Landgrebe 2015).

Local Indian soil. Soil samples from Maharashtra, Gujarat and Hisar (Haryana), provided by the NBSS, were imaged in the laboratory with a Resonon Pika NIR-320 line-scan camera (320 spatial pixels, 168 bands, 888.7–1714.6 nm). We use the Black, Red and Yellow soil folders (36 images). Each image shows six portions of one sample on a bright tray. Values are raw digital numbers: no white or dark reference frames are available. Soil type is confounded with acquisition session and moisture: all Black-soil images were acquired moist, all Yellow-soil images dry on one date, and Red soil both dry (one date) and moist. Moist and dry Red images share sample numbers, so images with the same number are treated as one sample (22 samples: 7 Black, 8 Red, 7 Yellow) in the soil-only protocols.

3. Methods

3.1. *Models and training*

LightGBM (Ke et al. 2017), a **random forest** (RF, 500 trees) and an RBF **SVM** use the centre-pixel spectrum (LightGBM: 100 trees, `max_delta_step`= 1, which prevented divergence on one Salinas split). All deep models receive a 15×15 patch. The **spectral transformer** sees only the centre pixel, one token per band (128-d, 4 layers, 4 heads). The **grouped-band transformer**, inspired by SpectralFormer (Hong et al. 2022) but without its cross-region fusion or patch-wise input, also sees only the centre pixel, with four adjacent bands per token (64-d, 4 layers); bands that do not fill the last group are zero-padded (Pavia University), except in the water-band-removed soil variant, where the last 3 of 119 bands are dropped. The **conv-stem transformer** is the spectral transformer with band tokens produced by two 3-D convolutions over the patch. **HybridSN** (Roy et al. 2020) has three 3-D convolutions, one 2-D convolution and two dense layers. All deep models use AdamW (learning rate 5×10^{-4}, weight decay 10^{-4}), cosine annealing, batch size 64 and at most 50 epochs with early stopping (patience 10) on validation loss. Inputs are standardised per band with training statistics. Runs used PyTorch 2.11, CUDA 12.8 and LightGBM 4.6 on Kaggle (NVIDIA Tesla T4 for the deep models).

3.2. *Protocols*

Random pixels (Indian Pines, leaky). For budgets $p \in \{1, 5, 10, 20\}\%$, p of the labelled pixels are drawn by stratified sampling, 10% of the rest form the validation set and the remainder the test set. Training and test windows overlap.

Disjoint tiles (all scenes, controlled). Each scene is divided into square tiles (24 pixels for Indian Pines, 32 for the others), half of which form the training area. For each seed, the best of 400 random assignments is kept, ranked by the number of classes with $\geq$25% of their pixels in the training area and $\geq$10% in the test area. Test pixels within 7 pixels of a training tile are dropped, so no test window contains a training-area pixel. Training pixels are drawn from the training area in proportion to its class frequencies, to a total of p of all labelled pixels ($\{1, 5, 10, 20\}\%$ for Indian Pines, $\{0.5, 1, 2, 5\}\%$ for the larger scenes); up to 3 000 other training-area pixels form the validation set. Only classes present in the training area are scored: 10–13 of 16 on Indian Pines, all classes on the other scenes.

Soil patches. Images of one sample are never split. In each of five repeats, the samples of each class are shuffled; 2 go to the test set, 1 to validation, and the budget $k \in \{1, 2, 3, 4\}$ is the first k remaining samples per class. At most 300 patches per training or validation sample are used. *Raw*: every non-overlapping 15×15 patch (the protocol of most HSI pipelines). *Soil-only*: a soil mask is computed per image (band-mean image divided by its column-wise 90th percentile, which removes vignetting; soil where the ratio is below 0.6), and only patches with $\geq$90% soil are kept. *Normalised*: soil-only patches with each spectrum divided by its mean over bands, which removes overall brightness.

Leakage checks. Tray-only: LightGBM and SVM are trained and tested on patches with $\leq$10% soil from the same samples (4 samples per class). *Cross-state*: Red is the only class imaged both dry and moist; models are trained on Black, Yellow and one Red state and tested on the other Red state, with or without the water bands (932–1010 and 1344–1498 nm) or brightness normalisation.

Table 1. Indian Pines overall accuracy (%) under a random pixel split and spatially disjoint tiles (mean ± SD over 5 seeds).

Split	Labelled	LightGBM	Spectral tr.	Grouped-band tr.	Conv-stem tr.	HybridSN
Random pixels	1%	51.01 ± 2.29	46.69 ± 2.11	46.48 ± 1.68[a]	39.53 ± 1.75[a]	**53.19 ± 2.28**
	5%	69.81 ± 0.64	61.62 ± 1.03[a]	58.89 ± 2.20[a]	72.55 ± 5.82	**89.01 ± 1.16**[a]
	10%	76.58 ± 0.85	74.64 ± 1.53	66.37 ± 1.26[a]	90.25 ± 2.38[a]	**97.02 ± 0.47**[a]
	20%	82.55 ± 0.59	76.84 ± 2.12[a]	69.64 ± 1.35[a]	92.10 ± 2.99[a]	**99.20 ± 0.23**[a]
Disjoint tiles	1%	**37.38 ± 6.49**	32.90 ± 9.14	35.15 ± 9.97	33.01 ± 9.29	34.95 ± 7.86
	5%	**46.90 ± 5.31**	37.57 ± 7.71[a]	35.56 ± 8.67[a]	31.10 ± 16.36	42.26 ± 7.84
	10%	**46.84 ± 5.56**	42.52 ± 5.98	40.03 ± 6.57	37.89 ± 10.12	43.25 ± 10.61
	20%	48.99 ± 6.23	**51.36 ± 5.56**	48.03 ± 5.24	40.95 ± 10.70	45.16 ± 6.43

tr. = transformer. [a]Paired difference from LightGBM significant after BH-FDR over the 100 airborne tests ($q < 0.05$).

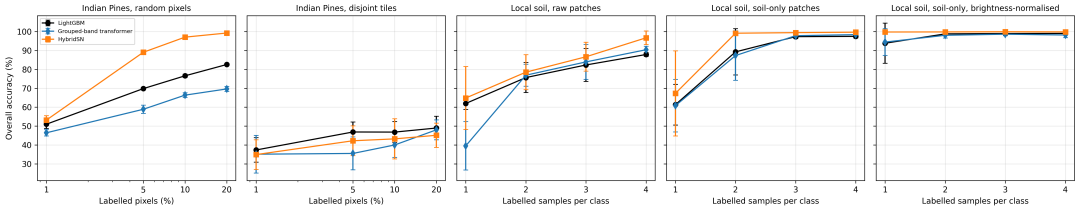


Figure 2. Overall accuracy of LightGBM, the grouped-band transformer and HybridSN under leaky and controlled protocols: Indian Pines with random pixels and disjoint tiles (left two panels), and Local soil with raw, soil-only and brightness-normalised patches (right three panels). Mean ± SD over 5 seeds.

Statistics. For each model we report the mean paired difference from LightGBM over the five seeds, its 95% confidence interval and the paired t-test p-value adjusted with the Benjamini–Hochberg procedure (q). On the airborne scenes HybridSN is also compared with the random forest, the strongest classical baseline. We correct within two families: every test on the airborne scenes (both split protocols, 100 tests), and every model-versus-LightGBM test on the soil data (all three patch protocols, 36 tests).

4. Results

4.1. *Spatial leakage*

Table 1 and Figure 2 (left) compare the two splits of Indian Pines. With random pixels, HybridSN is the most accurate model at every budget and 16.6–20.4 pp ahead of LightGBM from 5% ($q < 0.001$); the conv-stem transformer is 9.6–13.7 pp ahead from 10% ($q \leq 0.010$). With disjoint tiles, accuracy drops to 31–51% and the ranking changes: LightGBM has the highest mean at 1–10%, no deep model is significantly better than it at any budget, the two centre-pixel transformers are significantly worse at 5% (−9.3 and −11.3 pp, $q \leq 0.025$), and at 1% the random forest (41.3%) is significantly more accurate than HybridSN (+6.4 pp, $q = 0.020$). HybridSN falls from 99.2% to 45.2% at 20%. Part of the drop reflects a harder test set (classes that cannot be split are not scored, and test pixels are spatially separated), but the reversal of the ranking is the effect of the protocol.

Table 2. Overall accuracy (%) on spatially disjoint test areas (mean ± SD over 5 seeds).

Scene	Labelled	LightGBM	RF	SVM	Spectral tr.	Grouped-band tr.	Conv-stem tr.	HybridSN
Pavia Univ.	0.5%	72.34 ± 3.97	75.03 ± 3.55	75.10 ± 3.21	70.69 ± 2.85	69.52 ± 2.86[a]	74.78 ± 5.12	**80.71 ± 5.60**[a,b]
	1%	74.22 ± 5.83	74.87 ± 5.74	74.43 ± 4.62	72.04 ± 6.56	71.78 ± 6.09	79.90 ± 4.70	**83.36 ± 3.03**[b]
	2%	79.15 ± 2.98	77.66 ± 1.95	75.18 ± 2.73[a]	77.69 ± 5.24	73.57 ± 4.47[a]	85.94 ± 4.95[a]	**86.01 ± 5.53**[b]
	5%	83.04 ± 3.57	81.46 ± 3.19	78.27 ± 3.14[a]	83.21 ± 5.07	79.37 ± 3.33[a]	87.36 ± 4.99	**90.17 ± 4.41**[a,b]
Salinas	0.5%	73.59 ± 2.97	76.41 ± 3.11[a]	64.46 ± 10.51	64.97 ± 7.18	69.55 ± 5.77	70.28 ± 2.96	**76.67 ± 2.55**
	1%	77.77 ± 3.08	79.00 ± 2.76[a]	70.17 ± 4.80[a]	72.77 ± 3.63	74.92 ± 2.42	79.85 ± 3.83	**85.97 ± 3.00**[a,b]
	2%	80.30 ± 1.47	81.53 ± 1.54	75.57 ± 3.57[a]	78.13 ± 3.99	78.02 ± 0.87	83.04 ± 1.21[a]	**87.19 ± 3.36**[a,b]
	5%	81.91 ± 1.79	82.27 ± 2.49	79.05 ± 2.64	81.49 ± 2.33	81.28 ± 2.46	86.59 ± 1.14[a]	**89.03 ± 1.22**[a,b]

RF = random forest. Test areas contain 16 173–17 500 (Pavia University) and 17 802–18 986 (Salinas) pixels, depending on the seed. [a]Paired difference from LightGBM significant ($q < 0.05$, as in Table 1). [b]HybridSN significantly ahead of the random forest ($q < 0.05$).

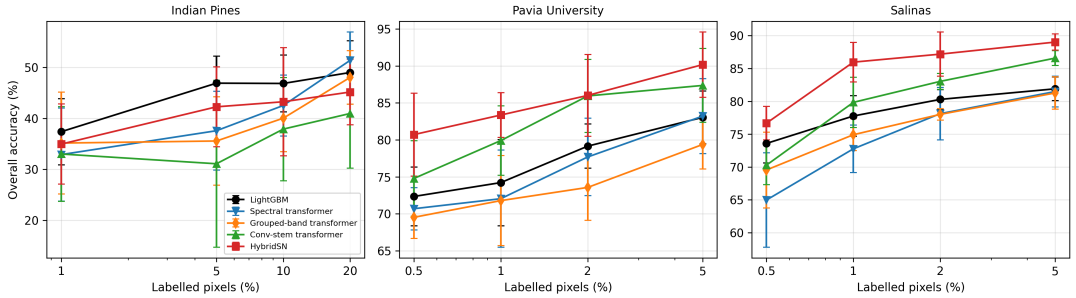


Figure 3. Overall accuracy on spatially disjoint test areas against the labelled share (mean ± SD over 5 seeds, logarithmic axis).

4.2. *Disjoint benchmark on three scenes*

Under disjoint tiles the advantage of patch-based models depends on the scene (Table 2, Figure 3). On Pavia University and Salinas, HybridSN is the most accurate model at every budget, 6.9–9.1 pp above LightGBM on Pavia University and 3.1–8.2 pp on Salinas (significant in five of eight cases, $q \leq 0.025$). It is also significantly ahead of the random forest at every budget on Pavia University (+5.7 to +8.7 pp) and from 1% on Salinas (+5.7 to +7.0 pp), but level with it at 0.5% on Salinas (+0.3 pp). The random forest matches or exceeds LightGBM at the smallest budgets (significantly on Salinas at 0.5% and 1%), and the SVM is the weakest classical model. The conv-stem transformer is significantly ahead of LightGBM at 2% on both scenes and at 5% on Salinas. The centre-pixel transformers are never significantly better than LightGBM; the grouped-band transformer is significantly worse on Pavia University at 0.5%, 2% and 5%. Average accuracy and Cohen's κ give the same best model in every row except Salinas at 0.5%, where LightGBM has a slightly higher AA (Appendix, Table A1).

4.3. *Acquisition leakage in laboratory soil images*

In the soil images 38–72% of the non-overlapping patches contain ≤10% soil (measured on 24 images). Trained and tested only on such tray patches, LightGBM reaches 87.1 ± 2.0% and the SVM 78.2 ± 2.3% (chance 33%): the tray identifies the imaging session, and with it the class.

The patch protocol changes both accuracy and ranking (Table 3, Figure 2, right). With raw patches, accuracy rises slowly with the budget, HybridSN is significantly ahead of LightGBM with four samples per class (+8.8 pp, $q = 0.022$), and the SVM is significantly worse than LightGBM at every budget (−6.6 to −11.5 pp). With soil-only

Table 3. Local soil overall accuracy (%) on held-out samples under three patch protocols (mean ± SD over 5 repeated sample-level splits).

Patches	Samples/class	LightGBM	SVM	Grouped-band tr.	HybridSN
Raw	1	61.97 ± 3.14	50.51 ± 4.23^{a}	39.56 ± 12.81	$\mathbf{64.85 \pm 16.63}$
	2	75.66 ± 7.90	64.36 ± 6.17^{a}	76.90 ± 5.79	$\mathbf{78.50 \pm 9.27}$
	3	82.31 ± 8.67	74.79 ± 9.10^{a}	83.93 ± 9.24	$\mathbf{86.64 \pm 7.64}$
	4	87.84 ± 1.01	81.28 ± 2.37^{a}	90.36 ± 1.86	$\mathbf{96.67 \pm 3.57}^{a}$
Soil-only	1	61.34 ± 10.78	$\mathbf{67.78 \pm 10.33}$	60.81 ± 13.90	67.32 ± 22.49
	2	89.24 ± 12.25	88.61 ± 12.74	87.30 ± 13.06	$\mathbf{99.10 \pm 0.89}$
	3	97.29 ± 1.09	98.43 ± 0.44	97.79 ± 0.86	$\mathbf{99.40 \pm 0.09}$
	4	97.44 ± 0.95	98.55 ± 0.50^{a}	98.35 ± 0.73	$\mathbf{99.64 \pm 0.16}^{a}$
Soil-only, normalised	1	93.79 ± 10.60	93.60 ± 10.25	94.41 ± 7.05	$\mathbf{99.73 \pm 0.16}$
	2	98.75 ± 0.31	99.04 ± 0.38	98.00 ± 1.44	$\mathbf{99.78 \pm 0.16}^{a}$
	3	98.93 ± 0.60	99.18 ± 0.39	98.57 ± 0.79	$\mathbf{99.88 \pm 0.09}$
	4	98.98 ± 0.54	99.13 ± 0.42	98.15 ± 1.42	$\mathbf{99.80 \pm 0.19}$

Raw: all patches, 24 samples (Red moist and dry counted separately), 5 460–5 628 test patches. Soil-only: 22 samples (same-numbered Red images merged), 1 531–3 686 test patches. [a]Paired difference from LightGBM significant after BH-FDR over the 36 soil tests ($q < 0.05$).

Table 4. Cross-state test: share (%) of Red test patches predicted Red / Black when the other Red moisture state is in training (mean over 5 seeds; soil-only patches).

Red test state	Model	All bands	Water bands removed	Normalised
Moist	LightGBM	97 / 3	89 / 10	100 / 0
	SVM	99 / 1	99 / 1	99 / 1
	Grouped-band tr.	97 / 3	96 / 4	99 / 1
	HybridSN	100 / 0	100 / 0	100 / 0
Dry	LightGBM	42 / 56	28 / 69	93 / 7
	SVM	56 / 43	34 / 65	93 / 7
	Grouped-band tr.	56 / 44	33 / 66	90 / 7
	HybridSN	71 / 29	49 / 51	98 / 2

patches every model exceeds 97% from three samples per class; HybridSN's lead shrinks to +2.2 pp, and the SVM is now slightly ahead of LightGBM (+1.1 pp, $q = 0.039$). After brightness normalisation every model reaches at least 93.6% with one sample per class and 98% with two, and the only remaining significant difference is HybridSN at two samples per class (+1.0 pp, $q = 0.005$).

Brightness, more than moisture absorption, links class to session (Table 4). With dry Red in training, moist Red is still recognised (89–100%). With moist Red in training, 29–56% of dry-Red patches are called Black, the other class imaged moist. Removing the water bands makes this worse (51–69% Black), whereas brightness normalisation largely removes it (90–98% Red). The soil types are therefore separable by spectral shape across moisture states, at least for Red, while absolute brightness carries session information.

4.4. *Band importance*

Exact TreeSHAP values (Lundberg et al. 2020) for LightGBM, plotted at the band centres from the sensor metadata (Figure 4), show that on soil-only patches one band dominates: 1414 nm, on the first overtone of the O–H stretch of water and hydroxyl, with 3.7 times the importance of the next band (1176 nm). Because Black soil was imaged only moist and Yellow only dry, this weight may reflect moisture rather than mineralogy. On Indian Pines (5% labelled, disjoint split) the top bands are at 400 nm

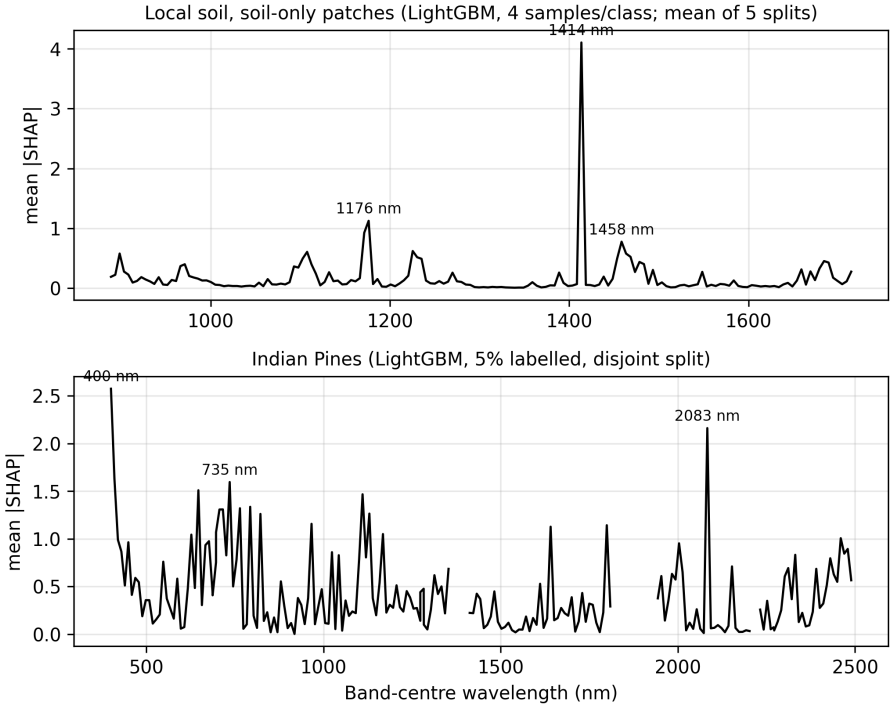


Figure 4. Mean absolute TreeSHAP value per band for LightGBM (summed over classes) on held-out test sets, against band-centre wavelength from the sensor metadata. Gaps mark removed water-absorption bands.

(the first band, where noise is high), 735 nm (the red edge of vegetation) and 2083 nm.

5. Discussion

The protocol can reverse the conclusion. On Indian Pines the random split supports the familiar claim that 3-D CNNs outperform gradient boosting by about 20 pp; the disjoint split does not. On the soil images, raw patches suggest that HybridSN pays off at larger budgets and that the SVM is a poor choice; once the tray is removed and brightness normalised, all learners are within about one point of each other from two samples per class. Neither leak is specific to one model: spatial leakage favours patch-based models, while acquisition leakage rewards any model that can read session cues.

What survives controlled evaluation. Models that convolve over the patch (HybridSN and the conv-stem transformer) beat LightGBM on Pavia University and Salinas under disjoint tiles, and HybridSN also beats the stronger random forest in seven of eight cases, consistent with the value of a local inductive bias (Hassani et al. 2021). Transformers that see only the centre spectrum never beat LightGBM significantly, consistent with the strength of tree ensembles on per-pixel input (Grinsztajn, Oyallon, and Varoquaux 2022).

Checklist. Before ranking models with few labels: (1) split spatially, with a buffer at least half the patch size; (2) for laboratory images, mask the background and report the share of background patches; (3) train a classifier on background-only patches, and treat above-chance accuracy as leakage; (4) report whether class is confounded with acquisition session, and test across sessions where any class allows it; (5) compare

models under brightness normalisation or radiometric calibration; (6) include more than one classical baseline, since a random forest outperformed LightGBM at the smallest budgets.

Limitations. The soil dataset has 22 samples, raw digital numbers and class confounded with session; only Red allows a cross-session test. Five seeds give wide intervals. All models use one training protocol without per-model tuning; on the airborne scenes 58–85% of deep runs reached the 50-epoch limit. On Indian Pines the disjoint split scores only the classes present in the training area.

6. Conclusion

The ranking of hyperspectral classifiers with few labels depended on the evaluation design as much as on the models. A random pixel split made HybridSN 17–20 pp better than LightGBM on Indian Pines from 5% labelled pixels; disjoint tiles removed the advantage, although HybridSN remained better on Pavia University and Salinas. In laboratory soil images, background and brightness cues let a tray-only classifier reach 87%; after masking and normalisation all models scored at least 94% with one labelled sample per class. Spatially disjoint splits, background masking and leakage checks should precede any claim that one model family needs fewer labels than another.

Disclosure statement

The authors report there are no competing interests to declare.

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Data availability statement

The Local Indian Soil images are available at `https://www.kaggle.com/datasets/dev123123456/local-soil-hyperspectral-dataset`; users should apply a soil mask (code provided), because most patches show only the sample tray. Indian Pines, Pavia University and Salinas are public benchmark scenes. Code, split definitions, soil masks, seeds, per-seed results, band-centre wavelength files and run manifests are available at `https://github.com/Devguru-codes/research_paper_code_and_results` (folder `revision/`).

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Appendix A. Average accuracy and Cohen's kappa

Table A1. Average accuracy (AA, %) / Cohen's κ on spatially disjoint test areas (mean over 5 seeds).

Scene	Labelled	LightGBM	Spectral tr.	Grouped-band tr.	Conv-stem tr.	HybridSN
Indian Pines	1%	**36.5** / **0.28**	29.2 / 0.23	31.7 / 0.26	20.7 / 0.19	31.6 / 0.25
	5%	**42.9** / **0.39**	34.6 / 0.28	33.2 / 0.26	28.7 / 0.22	38.4 / 0.33
	10%	**44.6** / **0.39**	40.5 / 0.33	39.7 / 0.31	32.4 / 0.27	40.3 / 0.34
	20%	47.3 / 0.41	**47.7** / **0.44**	44.9 / 0.40	37.0 / 0.31	43.0 / 0.36
Pavia Univ.	0.5%	64.6 / 0.58	61.9 / 0.55	58.5 / 0.53	53.8 / 0.59	**71.3** / **0.69**
	1%	68.2 / 0.61	64.6 / 0.57	63.9 / 0.57	68.6 / 0.68	**80.0** / **0.74**
	2%	73.5 / 0.68	73.9 / 0.66	70.3 / 0.60	80.9 / **0.78**	**82.8** / **0.78**
	5%	79.7 / 0.74	82.0 / 0.75	77.5 / 0.69	86.6 / 0.81	**89.1** / **0.85**
Salinas	0.5%	**79.3** / 0.70	68.9 / 0.61	73.7 / 0.66	65.9 / 0.66	79.0 / **0.73**
	1%	82.9 / 0.75	80.3 / 0.69	82.8 / 0.72	79.4 / 0.77	**89.8** / **0.84**
	2%	84.9 / 0.78	85.2 / 0.75	85.9 / 0.75	85.3 / 0.81	**91.4** / **0.85**
	5%	87.0 / 0.79	88.0 / 0.79	88.4 / 0.79	90.2 / 0.85	**93.0** / **0.88**

The best value per row is in bold.